

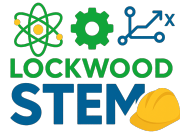
Introduction to Engineering Design

Aerospace Academy Version

Unit 0 Rubric

Introduction to Engineering Design

Course: Introduction to Engineering Design
Program: Academy of Advanced Aerospace Engineering
Brand: Lockwood STEM
Use: Student-facing and teacher grading rubrics



Unit 0

Course Launch, Procedures & FabLab Foundations

Course readiness, engineering notebook setup, digital workflow, FabLab safety, and mini challenge documentation.

Grading Rubrics and Scoring Guide

Unit 0: Course Launch, Procedures & FabLab Foundations

Assessment Purpose

Course readiness, engineering notebook setup, digital workflow, FabLab safety, and mini challenge documentation. This rubric is designed for consistent grading, feedback, and student self-checks.

Suggested 4-Point Scale

4 = Expert: exceeds expectations with clear evidence, precision, and independence. **3 = Proficient:** meets expectations with complete and accurate work. **2 = Emerging:** partially meets expectations; work is incomplete, vague, or inconsistent. **1 = Incomplete:** missing, unsafe, inaccurate, or shows limited evidence of understanding.

Unit 0 Score Summary

Category	Points Possible	Points Earned
Course readiness checklist	10	
Engineering notebook setup	15	
Digital workflow practice	10	
FabLab safety quiz/scenarios	20	
Safety agreement	10	
Mini challenge planning	10	
Mini challenge documentation	15	
Professionalism and cleanup	10	



Lockwood STEM

Academy of Advanced Aerospace Engineering

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Unit 0 Project Rubric



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete	
Course Readiness	10	All procedures are understood and checklist is complete; student demonstrates professional classroom routines independently.	Checklist is complete and student understands most routines.	Checklist is partially complete or student needs repeated reminders.	Checklist is missing or student does not demonstrate understanding of expectations.	Introduction to Engineering Design Aerospace Academy Version
Engineering Notebook Setup	15	Notebook is fully organized with title page, table of contents, numbered pages, entry format, and first entry completed with detail.	Notebook is mostly organized with minor missing elements.	Notebook is partially set up or lacks consistency.	Notebook is missing, disorganized, or unusable.	
Digital Workflow	10	Folders and files are correctly created, named, versioned, and submitted without support.	Folders and files are mostly correct with minor errors.	Folder structure or file naming has multiple issues.	Digital setup is missing or not submitted.	
FabLab Safety Understanding	20	Demonstrates strong understanding of PPE, permission, certification, emergency reporting, and cleanup through quiz and scenarios.	Demonstrates adequate understanding with minor gaps.	Shows partial understanding; reteaching is needed before lab access.	Unsafe or incomplete understanding; no lab access until remediated.	
Safety Agreement	10	Agreement is complete and student can explain why each expectation matters.	Agreement is complete.	Agreement is incomplete or missing required signature/initials.	Agreement is missing.	
Mini Challenge Planning	10	Team roles, criteria, constraints, sketches, and test plan are complete and thoughtful.	Planning is complete with adequate detail.	Planning is partially complete or vague.	Planning is missing or completed after building.	
Mini Challenge Documentation	15	Results, failure analysis, redesign notes, teamwork reflection, and cleanup reflection are complete and specific.	Documentation is complete with adequate detail.	Documentation is incomplete or lacks evidence.	Documentation is missing.	Rubrics
Professionalism & Cleanup	10	Consistently safe, respectful, productive, and leaves workspace fully reset.	Generally safe and productive with minor reminders.	Needs frequent reminders for behavior, safety, or cleanup.	Unsafe behavior or failure to reset workspace.	

Student Self-Check Before Submission

Requirement	Complete?
☐ Course readiness checklist evidence is complete and ready to be graded.	
☐ Engineering notebook setup evidence is complete and ready to be graded.	
☐ Digital workflow practice evidence is complete and ready to be graded.	
☐ FabLab safety quiz/scenarios evidence is complete and ready to be graded.	
☐ Safety agreement evidence is complete and ready to be graded.	
☐ Mini challenge planning evidence is complete and ready to be graded.	
☐ Mini challenge documentation evidence is complete and ready to be graded.	
☐ Professionalism and cleanup evidence is complete and ready to be graded.	
☐ All pages/files are named, organized, and submitted correctly.	
☐ Reflection or revision notes are complete.	

Teacher
Use the
evidence
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